Modules Information:	
Module	**HCI 3202 HUMAN COMPUTER INTERFACE**
Session	**JUN 2026**
Programme	**BITI: BACHELOR OF INFORMATION TECHNOLOGY**
Lecturer	**KALA DEVI MANAGURAN** kaladevi.managuran@newinti.edu.my Phone: 06 – 7982 000 ext.: - Room: 3A3-10

Summary of Coursework Breakdown:
(as stated in course structure)

No.	Description of Coursework	Learning Outcome Covered	Marks Allocated
1	Online Assessment	CLO2	20%
2	Group Assignment	CLO3	40%
	Contribution of the Coursework to the Course		**60%**
	Final Exam		**40%**
	Total		**100%**

Penalty for late submission:

After 3:30 p.m. – minus 20% of total mark awarded.
After 3:45 p.m. – minus 50% of total mark awarded.
After 4:00 p.m. – 0 mark for this piece of coursework

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The faculty reserves the right to use any means to identify plagiarism and copying. If you reference the work of other authors, you must ensure that you properly reference them to avoid the charge of plagiarism.

Copying the work of others (current students, past students, or the works of others) automatically earn you a Fail grade. If two or more students copied from each other, both students (regardless of who copies from whom) will earn a Fail grade.

ONLINE ACTIVITY			
Module	**HCI 3202 Human Computer Interface**		
Session	**JUN 2026**		
Programme	**BITI: BACHELOR OF INFORMATION TECHNOLOGY**		
Section	**6H1 / 6MB1**		
Lecturer	**KALA DEVI MANAGURAN**		
Coursework Type	**Online Assessment (Group)**		
Percentage	**20% out of 100%**		
Hand-out Date	**Week 5**	Due Date	**Week 7**

Students' Declaration:

I declare that:

- I understand what is meant by plagiarism.
- This assignment is all my own work, and I have acknowledged any use of the published or unpublished works of other people.
- I hold a copy of this assignment which I can produce if the original is lost or damaged.
- Attached are the links for the assignment submission:

Current Website Link	
Prototype Link	

[Name / ID] **ROWAN CESAR MACUACUA/ I25037217**

SHAHADAT HOSSAIN / I26041279

RADUAN MIA /I25035605

MORRIS KINYUA /I24028729 [Signature] _____

Learning Outcomes Accessed:

CLO2	Construct prototype interfaces of computer systems and various websites based on HCI design principles. (P4, PLO3)

Lecturer's Comments:
Penalty for Late Submission:
After 3:30 p.m. – minus 20% of total mark awarded. After 3:45 p.m. – minus 50% of total mark awarded. After 4:00 p.m. – 0 mark for this piece of coursework

Note: Plagiarism & Copying

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Copying the work of others (current students, past students, or the works of others) automatically earn you a Fail grade. If two or more students copied from each other, both students (regardless of who copies from whom) will earn a Fail grade.

ASSESSMENT CRITERIA AND GRADING SHEET

Task No.	*Breakdown of assessment criteria*	*Weight (%)*	*Range of achievement (✓)*					*Marks Awarded (%)*
			Excellent	*Above Average*	*Satisfactory*	*Below Average*	*Poor*	
			[20-17]	**[16-13]**	**[12-9]**	**[8-5]**	**[4-0]**	
1	Introduction	20						
2	Context of Current Websites Chosen	20						
3	Screenshot of Prototype Design and link to the prototype	20						
4	Justifications on Prototype Design based on Heuristic evaluation	20						
5	Conclusion and References	20						
TOTAL		**100%**						

ONLINE ASSESSMENT **(20%)**

You are required to design a prototype for a website below in a group of 3 to 4 members. You may use any free online tools such as Figma to design the prototype. Choose only one (1) website and make sure it is not same with others:

1) https://www.art.yale.edu/
2) http://irishwrecksonline.net/
3) https://www.mackido.com/
4) http://www.paperrad.org/oldindex2009.html
5) https://cloud9walkers.com/
6) https://www.discountbedsbelfast.co.uk/
7) https://www.mraireland.com/
8) https://dticket.railway.co.th/DTicketPublicWeb/home/Home
9) https://santapod.co.uk/index.php
10) https://ashendchildrensfarm.co.uk/

To do so, you must conduct HCI research related to the website that you choose above. You may also benchmark current related websites. **DO NOT refer to Wikipedia**. Perform appropriate **User Research**. Collect sufficient, accurate, and relevant information. Your report should have the following: Write up should not be more than **10 pages, excluding Assignment brief, cover page, table of contents and references**.

- Assignment Brief
- Project Cover Sheet
- Table of Contents:
    1. Introduction
    2. Context of Current Websites Chosen
    3. Screenshot of Prototype Design and link to the prototype
    4. Justification on Prototype Design based on Heuristic evaluation.
    5. Conclusion & References (Harvard Referencing system)
- Appendices:
    - Appendix A: Turnitin Report
    - Appendix B: Other supporting documents

FORMAT

Please refer and implement the report format guidelines below for this report:

a) Adhere to the following word processing requirements:
    - All heading should be capitalized, bold and left-aligned. Tab = 1cm
    - All paragraphs must be fully justified with 1.5 line spacing.
    - All text must be typed in font type “Times New Roman”, text size 12.
    - Include details for your name, student identification number, and assignment title on your front page.
    - Do not include unnecessary information such as logo, date, and assignment title in the header and footer section of each page. You are only allowed to insert page number in the bottom right footer section of each page.
b) Remember to spell check your report.
c) Use default margin settings.
d) Submit to Turnitin and make sure the percentage of plagiarism level < 20%.
    *Note: Exclude cover page, references, mark sheet when submitting to Turnitin.

ASSIGNMENT BRIEF

The purpose of this report is as a submission for an Online Assessment (20%) of the Unit: HCI 3202 Human Computer Interface, Learning Outcome: CLO2 Construct prototype interfaces of computer systems and various websites based on HCI design principles. Participants were to form a small group (3-6 people), select one web site from the given list, carry out Research with Users (RwU) and Human-Computer Interaction (HCI) studies of the Web site, and create a new prototype Web site interface with the above detailed findings from an online Web design tool. MRA Ireland's official website (https://www.mraireland.com/) has been selected for this project, as it is the official website for the governing body of Motocross, Quadcross, Sidecarcross, Enduro, Supermoto and Trials competition in Ulster, MRA Ireland (Motorcycle Racing Association (Ireland) Ltd).

The following parts of the report include an explanation of the context of the current website, usability problems noted during the comparatives and heuristic evaluation, the final design prototype and the reasons behind the key design choices according to usability heuristics. The updated prototype has been created in Figma and is available for sharing via Figma's shareable link for reference in Section 3 of this report.

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1. INTRODUCTION

Human-Computer Interaction (HCI) pertains to the design and evaluation of computing systems that interact with users, improving the systems being efficient, effective, safe and satisfying for the people who interact with them (Nielsen, 1994). Heuristic evaluation is a key technique in HCI for detecting usability problems in a given Web-based interface by systematically going through the interface with a small number of 'heuristics,' usability principles, to bring them to light prior to putting the interface before actual users. This report is made to the Motorcycle Racing Association (Ireland) Ltd, the governing body for off-road motorcycle disciplines, Motocross, Quadcross, Sidecarcross, Enduro, Supermoto and Trials within Ulster and this report applies to MRA Ireland (the official website of the Motorcycle Racing Association (Ireland) Ltd).

This assessment has two aims. The group first undertook HCI and comparative research in order to identify any usability problems with the existing MRA Ireland website for the key user groups – competitors, club officials and spectators who use the site to access information on the dates of events, or the results of championships or regulatory information. Second, the community utilized the results from this research to create a reengineered prototype user interface prototype in Figma that bolsters the subjected issues whilst maintaining the information content the neighborhood relies on.

The rest of this report is organized as follows: Section 2 outlines the background of the current website; why it exists, main user sections and the usability problems discovered during the inspection and competitor research. Section 3 shows the new prototype and the link to the interactive Figma prototype. Consequently, in section 4, we explain the important design choices we made in the prototype from the ten usability heuristics provided by Nielsen. The report is then concluded in Section 5 containing the sources used throughout, in Harvard referencing.

2. CONTEXT OF CURRENT WEBSITES CHOSEN

2.1 OVERVIEW OF THE WEBSITE

Built on the Wix platform, MRA Ireland (mraireland.com) is a news and information centre mainly to serve the off road motorcycling community of Ulster. It features a news feed on its homepage, an events calendar, five championship pages (Ulster Solo Adult, Ulster Quad & Sidecar, Ulster Solo Youth, Ulster Trial, and Irish Solo), a club directory, and a section on competitor information such as grading lists, the MRA rule book, number plate specifications, policies, and race licences. The main navigation includes a coaching page, an 'About' page, and a footer with the sponsors logo, which is visible at the bottom of each page.

2.2 PRIMARY USER GROUPS

- Players / Rides who must find out the next rounds, entry links and the rider's next ranking in the series.
- Officials and Organisers of the Club, who refer to the rule book, grading lists and licensing policies for conformance.
- People viewing the events and their families, accessing event dates and locations and general news.
- Visitors, which range from the casual to the avid fan, who are looking to view for the purposes of awareness and image.Visitors, including the casual fan, the avid fan, who come to view for the purpose of awareness and image.

2.3 USABILITY ISSUES IDENTIFIED

The co-iteration of the live site with Nielsen's usability heuristics and a review of how similar motor sports governing body / event administration sites present the same kind of information highlighted some common challenges:

Deep, text-heavy navigation: championship, competitor and coaching information is delivered via multi-level dropdown – several clicks to get to numerous pages of interest when we need one like the championship table or rule book.

- Fragmented event information is reproduced on home page in simple list format and on another page, the Events Calendar, information is provided on event dates but with varying detail (some rows have only the venue, and club, others have TBC); this makes it difficult to scan at a quick glance.

- Insufficient typography differentiation or spacing in the home page: news articles, the events list, and the sponsor strip vie for attention and very little differentiation in typography or spacing with the single thing that is most important in itself (the next round) not being immediately apparent.

- No search function at all; people aware of their search target (such as a specific rule / grading list can go through the entire menu structure instead of searching.

- Limited mobile affordances – interactive elements, like 'Enter Online' links are inconsistently rendered as buttons and common links, making recognisable when actions are clickable.

- External entry links route to third-party forms (e.g. Jotform, Google Forms) without contextual explanation, which can raise trust concerns for first-time users.

Analysing other sites in the motor sport and event-platform sphere (e.g. racing federation-specific event calendars) demonstrates that such sites are good models that make the calendar central and place it prominently at the front of the page, have consistent styling of links for register action, and generally organise competitor-related documents (rules, licences, grading) into a shallow information architecture (usually a flat, menu-style structure rather than a dropdown).

These patterns were used in a way to support the ideas behind the redesign priorities as outlined in Section 3, and as outlined previously in Section 4, these patterns were valid.

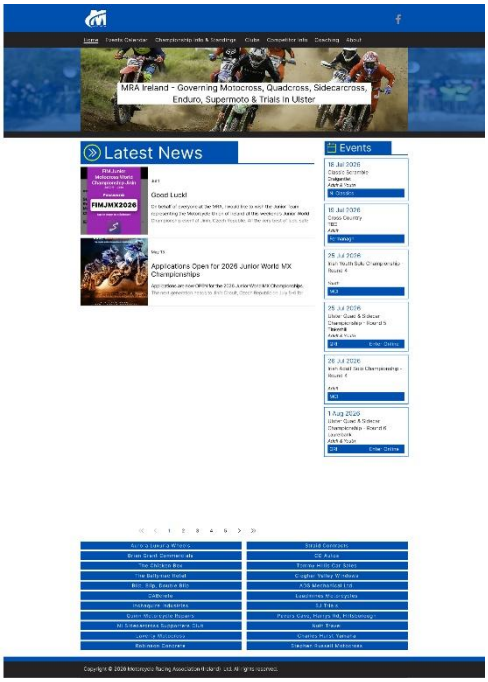
3. SCREENSHOT OF PROTOTYPE DESIGN AND LINK TO THE PROTOTYPE

The redesigned prototype was created in Figma. The interactive prototype, including all designed screens (homepage, events calendar, championship standings, and competitor information), can be accessed via the link below:

FIGMA MRA Website Design Project URL:
https://www.figma.com/design/uZiFj7QaebOuhsS1G9GNvP/HCI-MRA-Website-Design-Prototype?node-id=0-1

Screenshots of the key redesigned screens are presented below.

3.1 HOMEPAGE



3.2 EVENTS CALENDAR

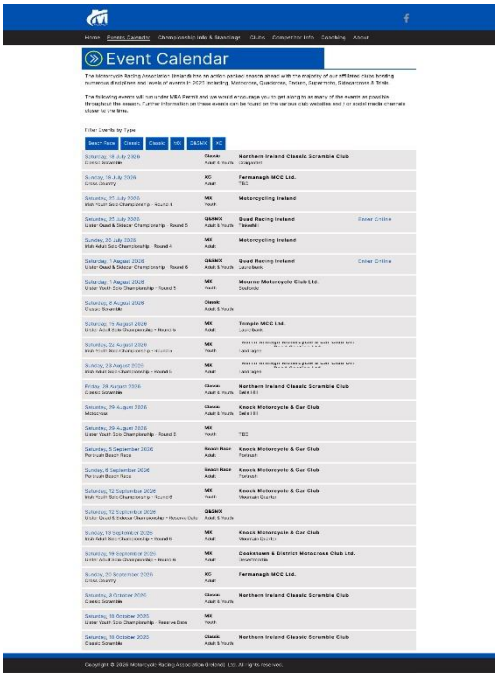


Figure 3.3 – Affiliated Clubs

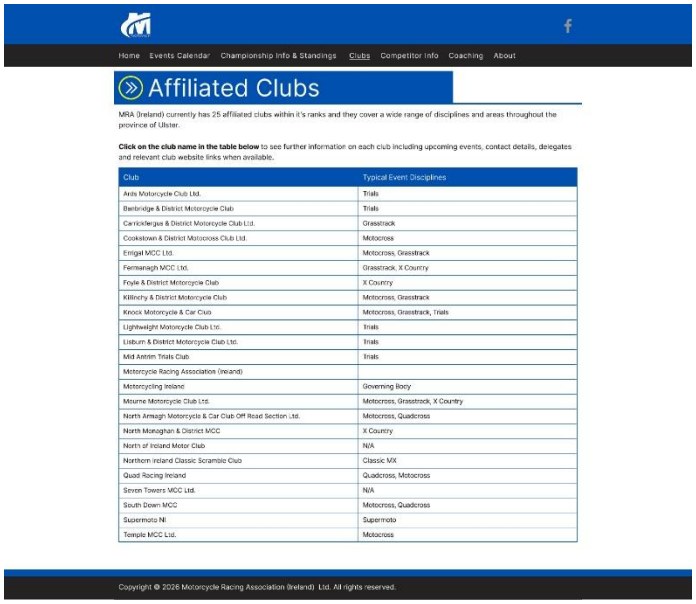


Figure 3.4 – Coaching

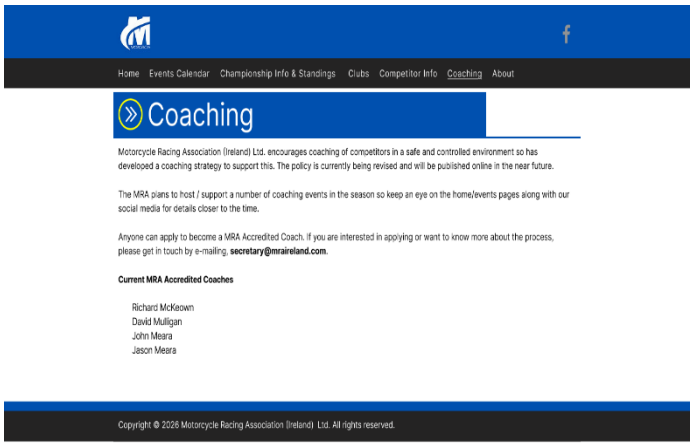


Figure 3.5 – About

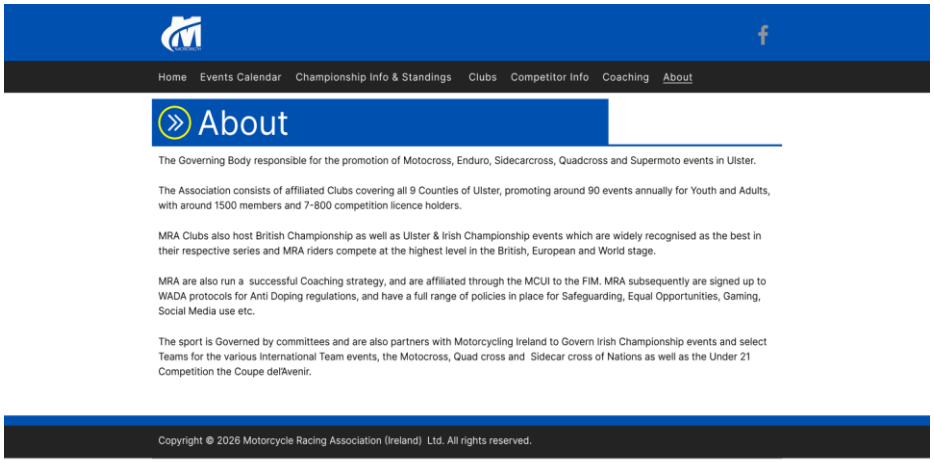


Figure 3.6 – Adult Solo Championshi

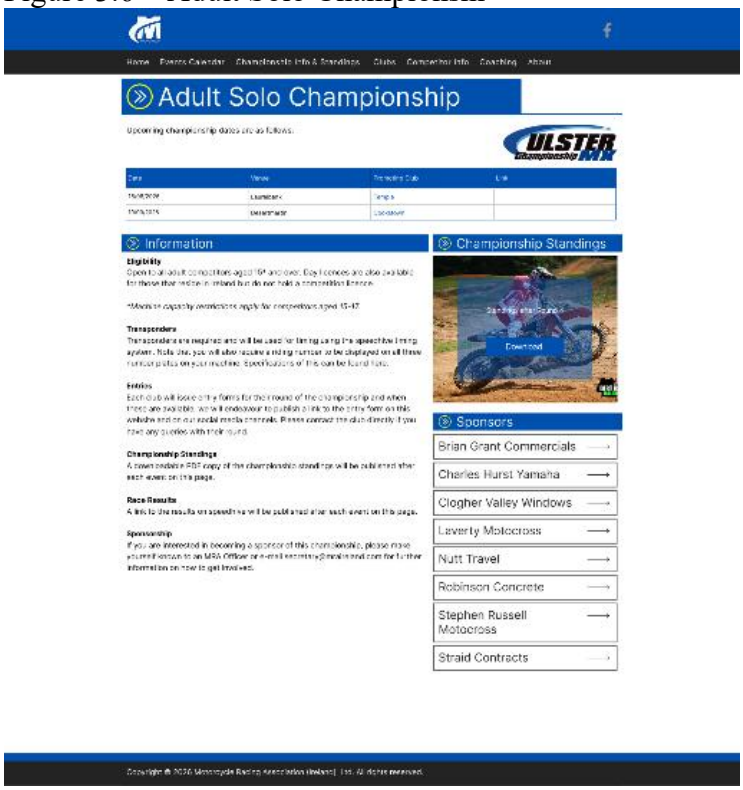


Figure 3.7 – Quad & Sidecar Championship

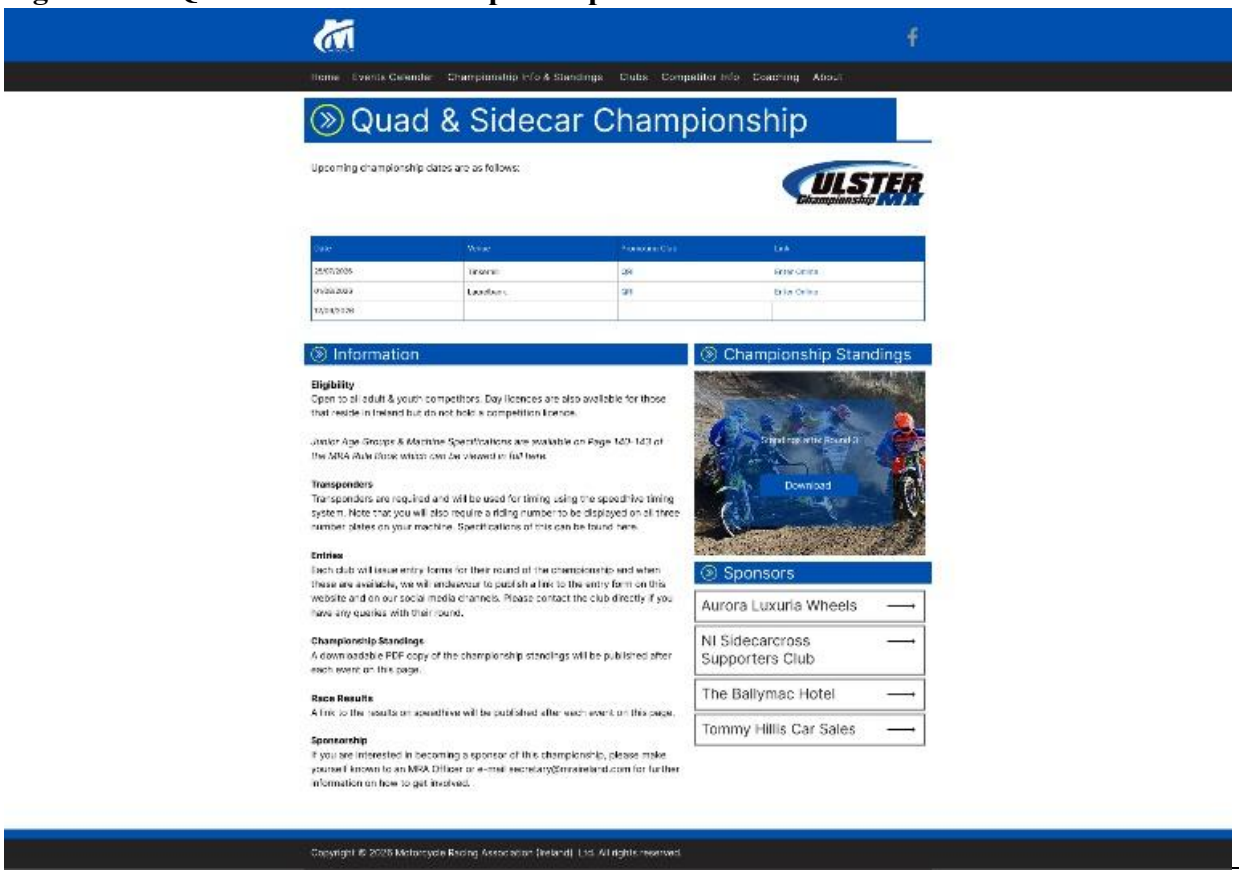


Figure 3.8 – Youth Solo Championship

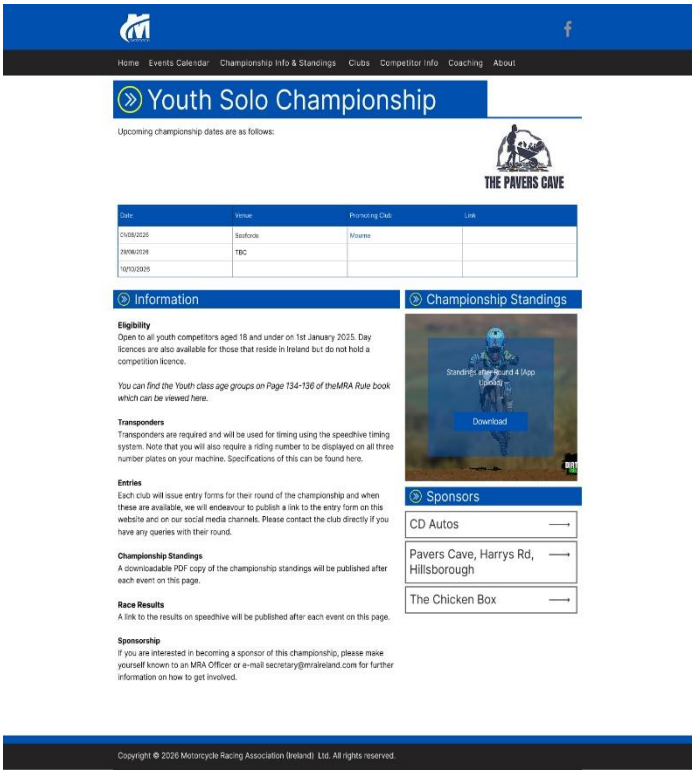


Figure 3.9 – Ulster Trial Championship

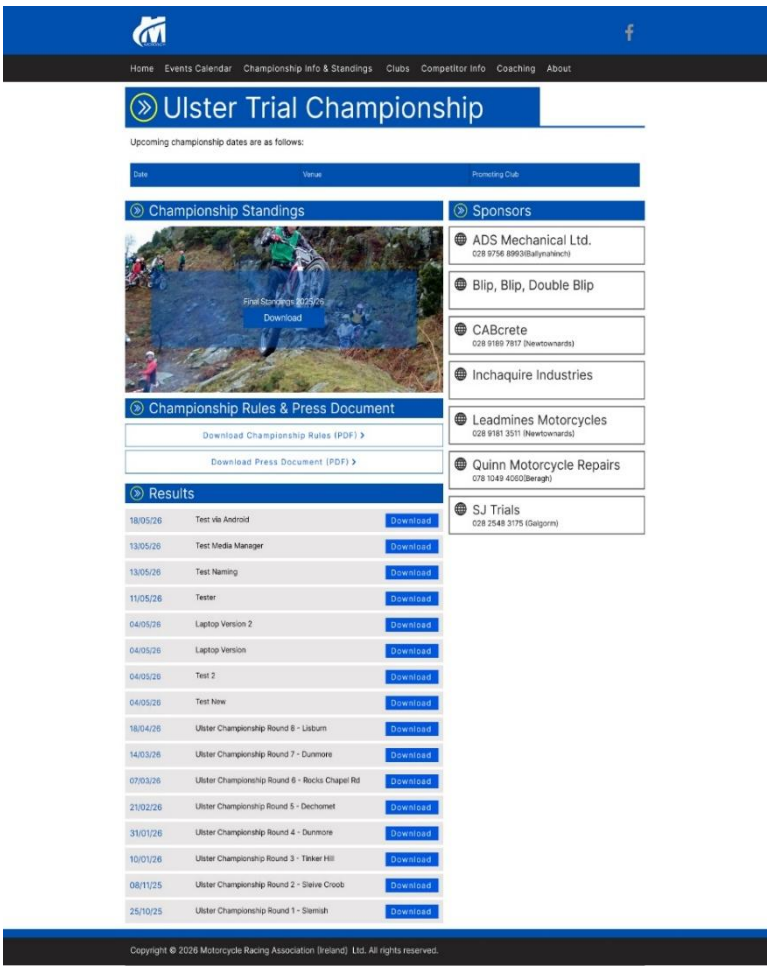


Figure 3.10 – Irish Solo Championship

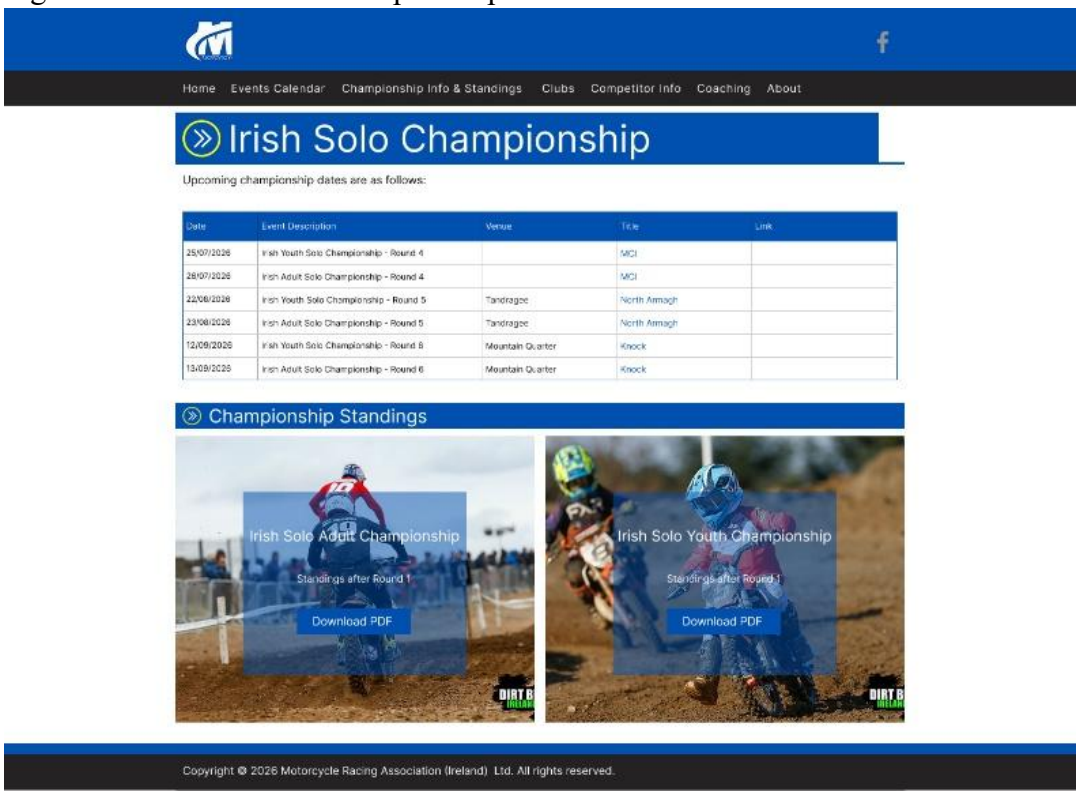


Figure 3.11 – Grading Lists

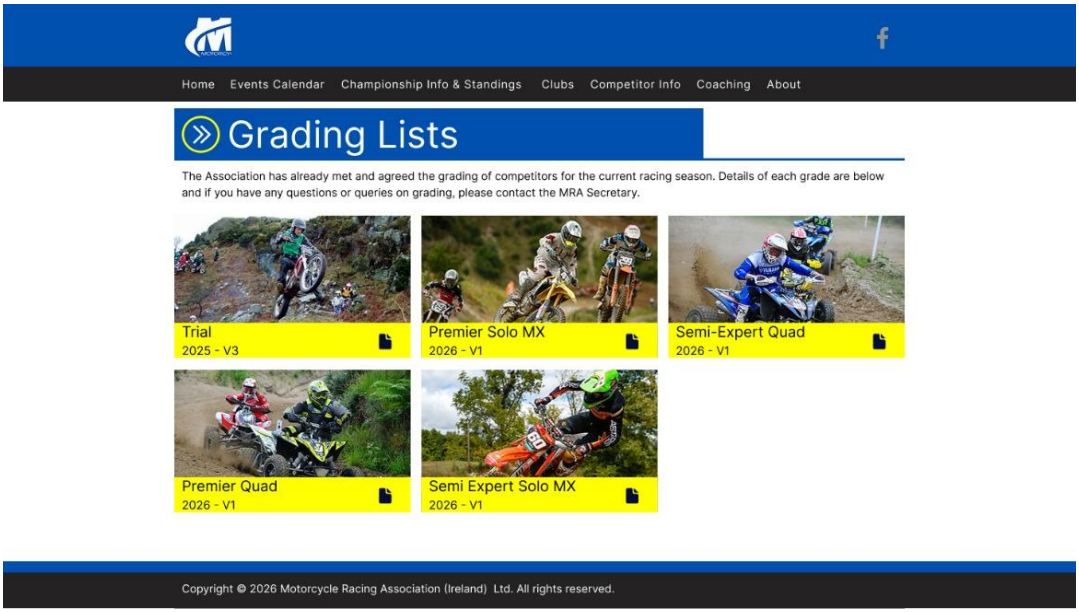


Figure 3.12 – Number Plate Specifications

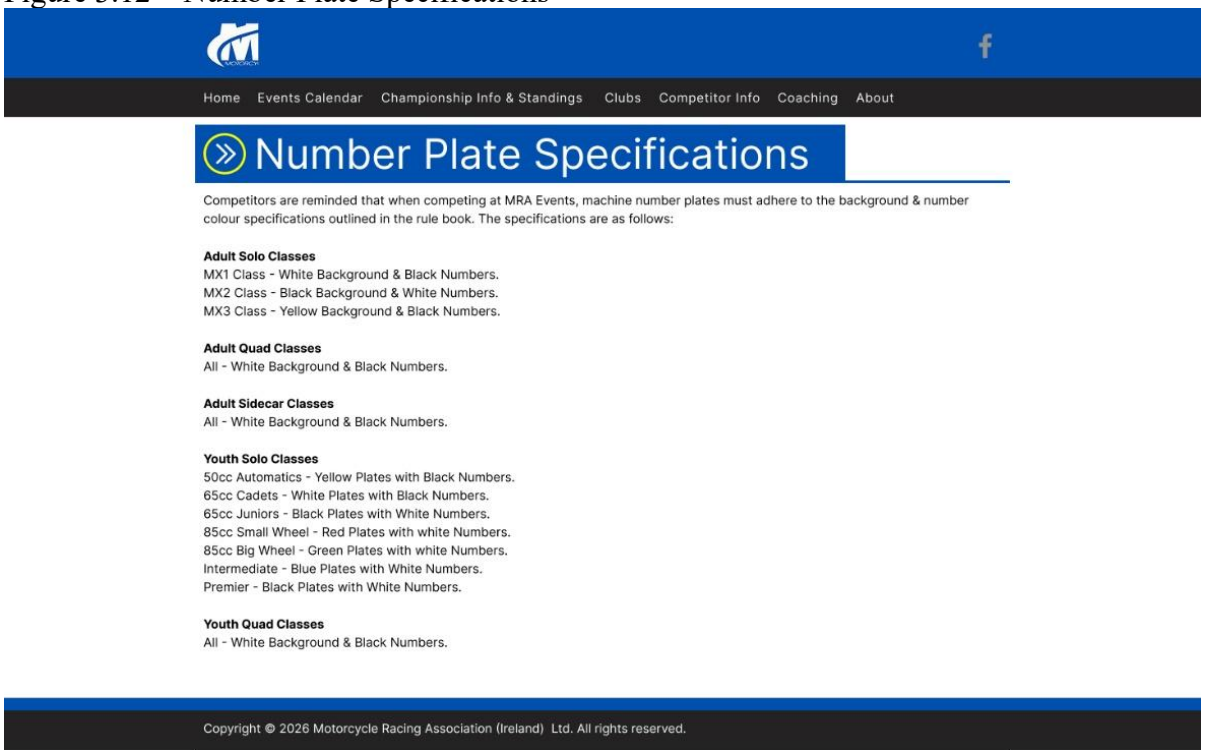


Figure 3.13 – Policies

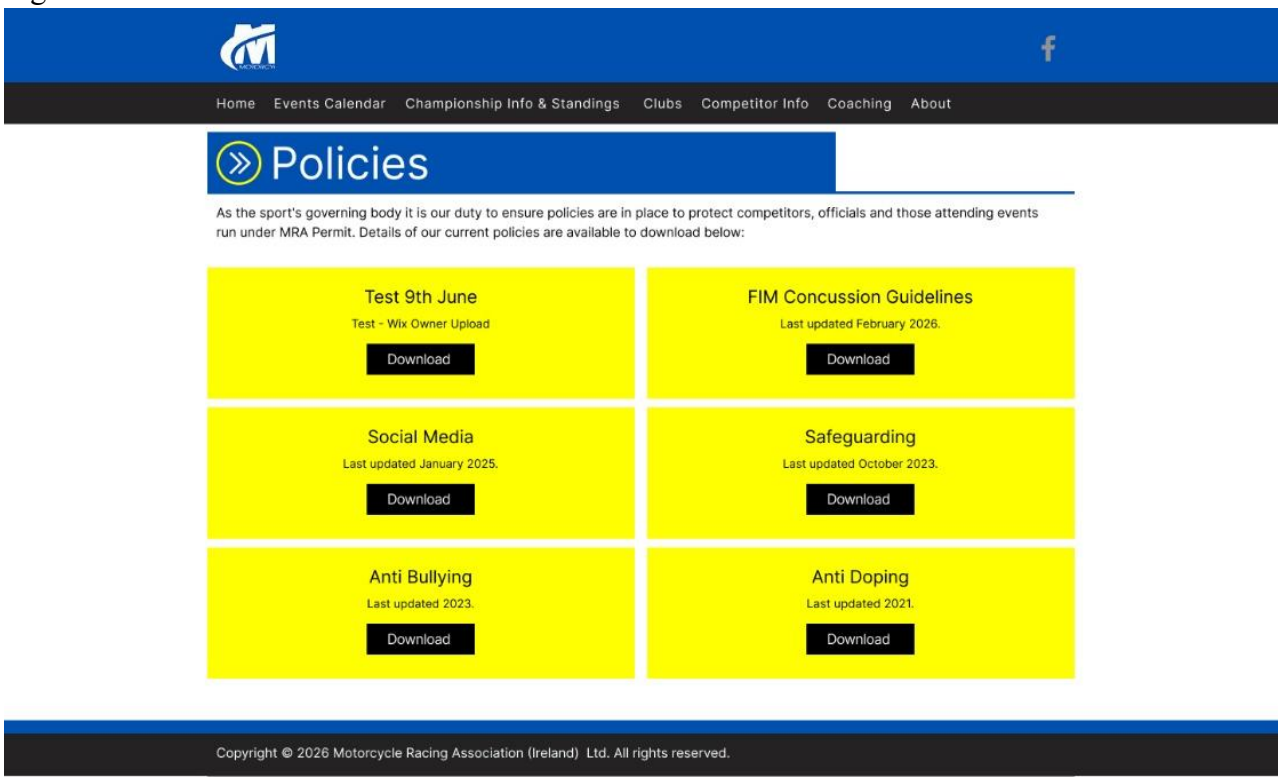


Figure 3.14 – Racing Outside Ireland

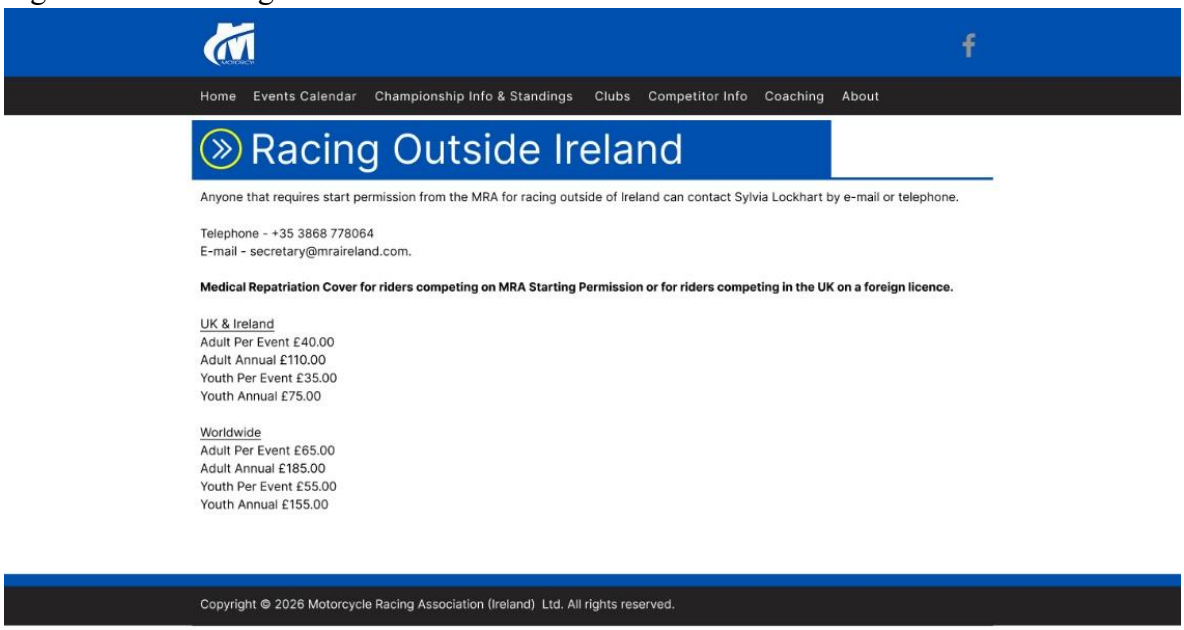
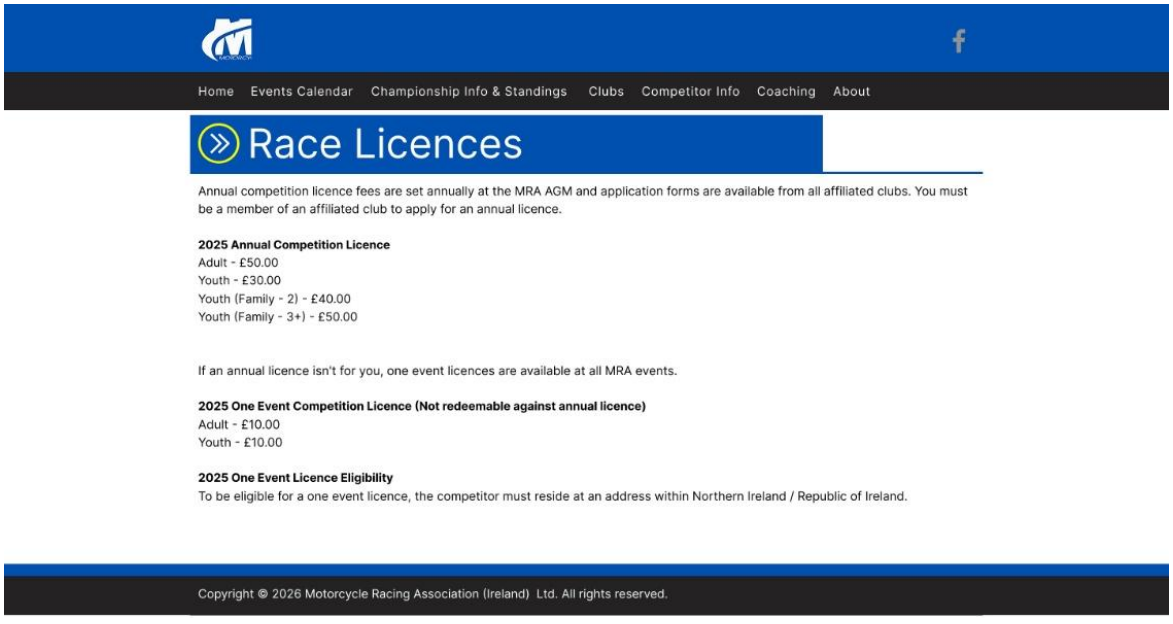


Figure 3.15 – Race Licenses



4. JUSTIFICATION ON PROTOTYPE DESIGN BASED ON HEURISTIC

This section explains why major design elements are used in the prototype that align with Nielsen's ten heuristics. These heuristics are a well-known set of criteria for assessing the understandability, consistency, efficiency and ease of use of an interface (Nielsen, 1994; Nielsen Norman Group, 2024).

4.1 VISIBILITY OF SYSTEM STATUS

The newly redesigned site home page has a distinctively labeled “Next Race” area, with a countdown along with event info specifies date, location and event name. It tells users right away of what the next event in time will be if they were to find out, without having to trawl through several pages. The Events Calendar also now provides details of the date, event category and venue of each fixture, allowing users to gain an understanding of the fixtures in the current fixture schedule and what information they have before selecting an event.

4.2 MATCH BETWEEN SYSTEM AND THE REAL WORLD

The prototype includes language that is easily understood by the motorcycle competition community- the competitors, officials and spectators. The separate Adult Solo Championship, Youth Solo Championship, Quad & Sidecar Championship are available, along with Grading Lists, Race Licences and Number Plate Specifications, and highlight the terminology already in use within motorcycle racing.

Information is also organised according to real racing activities. Championships are segmented by type of competition, licences, policies, coaching and grading documents are in clearly named sections. This will better equip the user with understanding of the website via their sport knowledge.

4.3 USER CONTROL AND FREEDOM

The prototype has a consistent navigation bar, and guides the user through the Home, Events Calendar, Clubs, Coaching and About sections. Other clearly labeled navigation elements lead to the information pages covering the championship and competers.

The users cannot, as a result, be locked in a single page, nor isolated from following only one path. They're able to switch back and forth to the general areas of the web site by clicking on choice of the navigation bar.

4.4 CONSISTENCY AND STANDARDS

The prototype has a similar layout across the site. Navigation Bar, typography, Color, button, spacing and footer look similar on pages.

Each of the championship pages (Adult Solo, Youth Solo, Quad & Sidecar, Ulster Trial and Irish Solo) have a similar page layout. This involves a big heading, introduction, championship intricacies, standings and sponsor information. By keeping the layout consistent, it can guide users to anticipate where information is likely to be located as they navigate from one championship page to another.

The buttons and links are displayed consistently as well. Buttons with commonly used roles are used to show actions to view documents, check standings or get more information instead of as body text.

4.5 ERROR PREVENTION

Prototype breaks information into pages and categories so the people who are familiar with the site will not be tempted to click some of the same alternate links they might have. For instance Adult Solo, Youth Solo, Quad & Sidecar, Ulster Trial and Irish Solo are not grouped together into one championship page, but are shown separately.

The Events Calendar also offers visible filters for the month and for the type of event and the category of the event's championship. These controls are used to narrow down the information displayed on the screen before selecting an event to prevent users from incorrectly opening events.

Document pages (Policies, Grading Lists and Race Licences) are titled descriptively and have clearly labelled action buttons. So there, users can see what the document is about, before clicking into it or downloading it.

4.6 RECOGNITION RATHER THAN RECALL

As soon as the user opens the prototype they will have the main options displayed, allowing them to not have to memorize where information is found. The navigation menu is not hidden and favourite categories are displayed with easy-to-read, clear labels, cards and buttons.

The available months and championships and event categories will be highlighted on the Events Calendar page with an available selection control. Standings, schedules and related information are also clearly displayed in championship pages within clearly differentiated sections. This decreases the amount of data the user needs to keep in their head as he or she is moving from one page to the next.

The familiar icons, document cards and descriptive headings further assist in recognition. For instance, users can recognize downloadable policies and grading documents at their title and the way they're displayed in the document.

4.7 FLEXIBILITY AND EFFICIENCY OF USE

The website has been redesigned to cater to different kinds of users. New participants will be able to use the home page and follow the main menu to other parts of the site, while the established competitor can navigate all the main pages of the site, including Race Licences, Grading Lists, Number Plate Specifications, or any specific championship.

The Events Calendar filters filter the number of events displayed based on the user's requirements. For instance, someone interested in Youth Solo events can checkout the event list and select the appropriate category, without having to read the entire list.

The homepage also has a direct link to future events and news. This minimises the number of steps for common information that has to be taken to access information.

4.8 AESTHETIC AND MINIMALIST DESIGN

The prototype clear visual hierarchy for organisation of information according to its importance. The next race is visually prominent on the homepage, followed by upcoming races, news and sponsor details.

Content is carded such that there are enough white spaces between content pieces and content parts. It helps prevent the entire page of text from appearing to be all in one big block, and makes the page more readable.

The palette of colours are mostly limited to dark blue, white and accent colours, which are part of the organisation's branding. Content that is less prominent is more subtle in visual emphasis while headings, buttons and other important content is emphasized. This results in a more streamlined interface, while keeping critical data available.

The redesign accordingly benefits the website through making it appealing and usable as well as readable and attention-getting and a means to task completion.

4.9 HELP USERS RECOGNISE, DIAGNOSE AND RECOVER FROM ERRORS

Even though there are no complex forms in the prototype, the navigation and structure of the content helps the users move away from mis-selections. Consistent navigation bar makes it easy for users to quickly get to another part of the navigation system if they accidentally click on another info or policy page or another championship.

Where headings are on the page, they will be highlighted to confirm which page heading you are on. For instance, they clearly mark pages as being for the Youth Solo Championship, Race Licences, Policies or Racing Outside Ireland etc. This enables users to identify easily when they've opened the incorrect page.

4.10 HELP AND DOCUMENTATION

Important information such as Race Licences, Policies, Grading Lists, Number Plate Specifications, Coaching, Racing Outside Ireland etc are available for specific pages of the prototype. Clearly defining the sections into which the documentation is structured helps users quickly and efficiently find what they are looking for.

5. CONCLUSION

This report analyzed the usability of a MRA Ireland website and provided a new prototype, created according to principles of human computer interaction (HCI). The new design will enhance user navigation, information organisation, visual hierarchy, and page consistency, while preserving some of the original characters, content and functions of the website.

The prototype is user-friendly and intuitive for the users (competitors, officials and viewers), as it adopts Nielsen's Ten Usability Heuristics which makes key information more easily accessible and understandable. Additional usability testing with actual users could be incorporated in future, to obtain feedback and make further improvements to the design.

REFERENCES

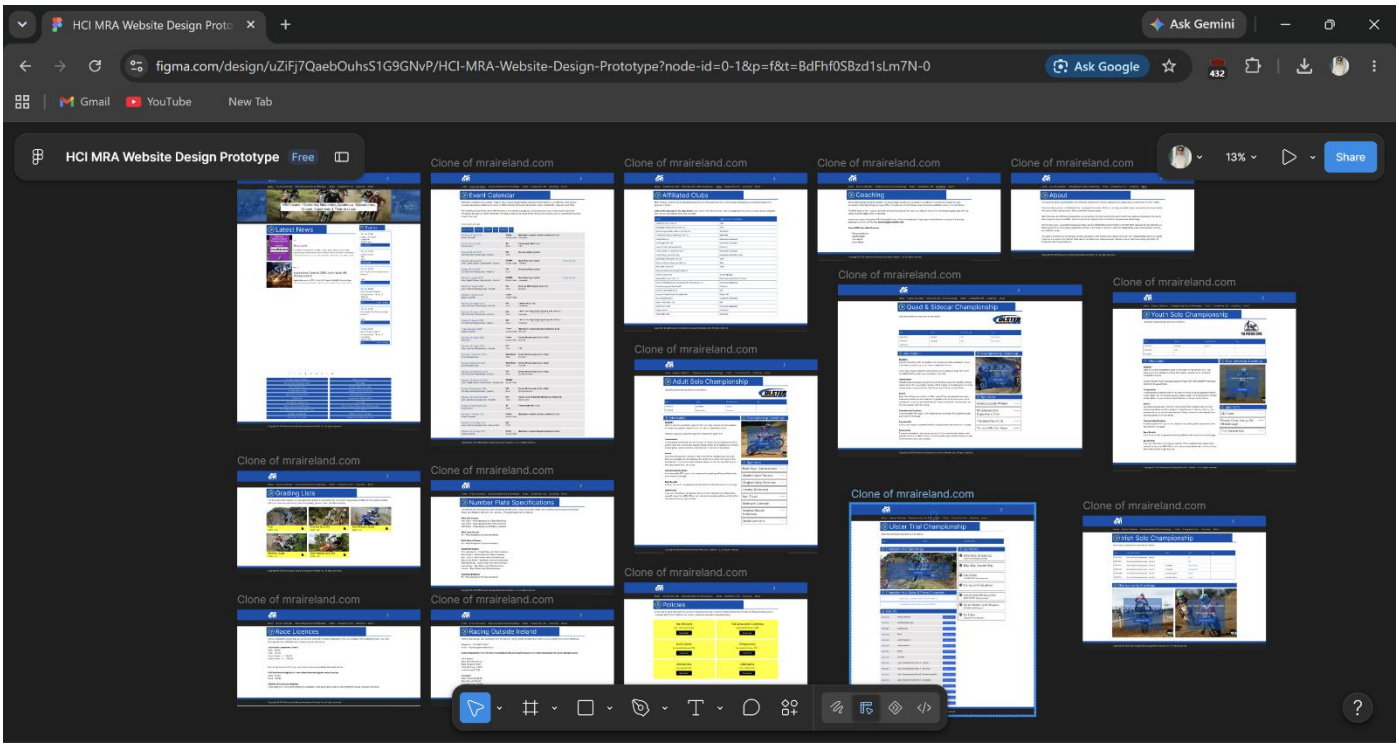
1. Nielsen, J. (1994) Usability Engineering. San Francisco: Morgan Kaufmann.
2. Nielsen Norman Group (2024) 10 Usability Heuristics for User Interface Design. Available at: https://www.nngroup.com/articles/ten-usability-heuristics/ (Accessed: 16 July 2026).
3. Nielsen, J. and Molich, R. (1990) 'Heuristic evaluation of user interfaces', Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, pp. 249–256.

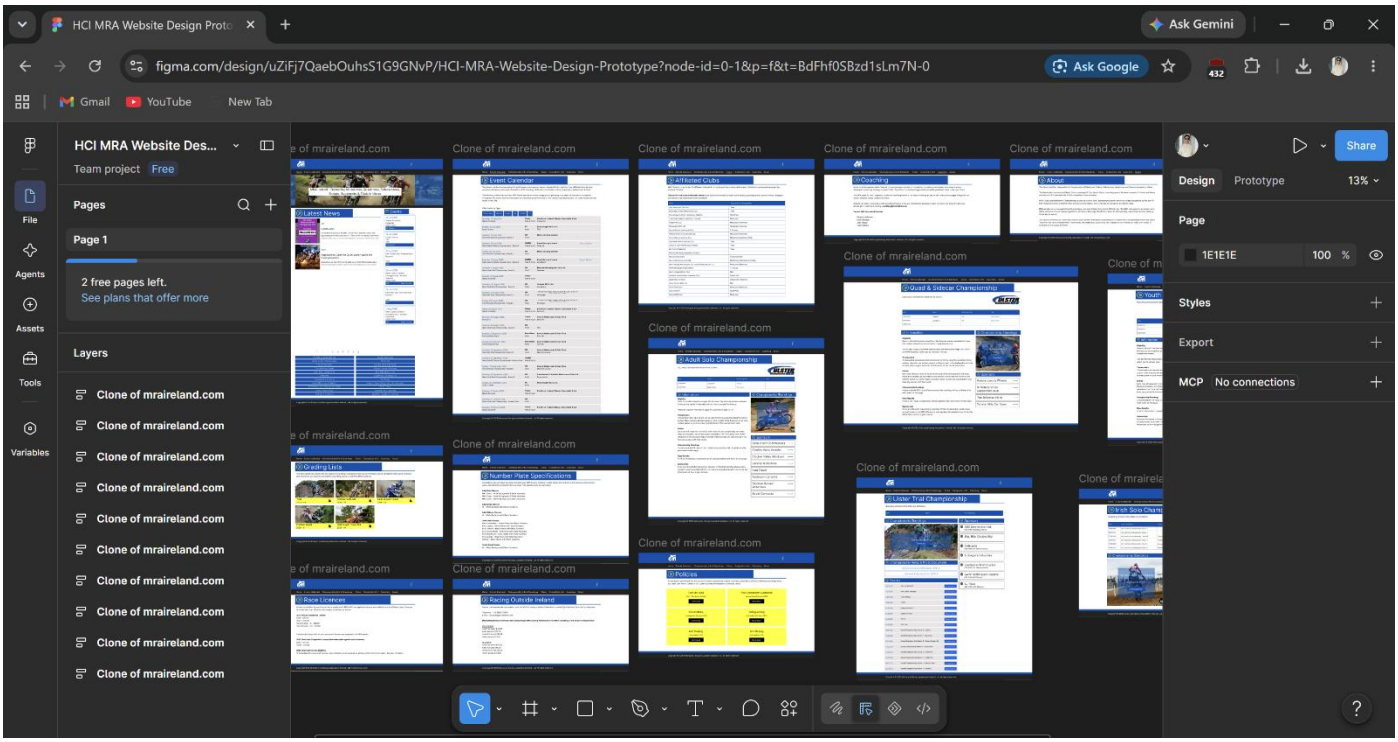
Appendices:

Group Contribution:

Type of Work	Shahadat Hossain (I26041279)	Rowan Cesar Macuacua (I25037217)	Raduan Mia (I25035605)	Morris Kinyua (I24028729)	Total
1. All 15 MRA Website Figma Designs	100%	0%	0%	0%	= 100%
2. Main Report Writing	0%	100%	0%	0%	= 100%
3. Report Formatting	50%	50%	0%	0%	= 100%

All 15 MRA Website Figma Design Screenshots as a Evidence of Conduct:




HCI MRA Website Design Prototype
figma.com/design/uZiFj7QaebOuhsS1G9GNvP/HCI-MRA-Website-Design-Prototype?node-id=0-1&p=f&t=BdFhf0SBzd1sLm7N-0
Ask Gemini
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